

CHAPTER TWO

INTELLIGENT AGENT

Intelligent Agent

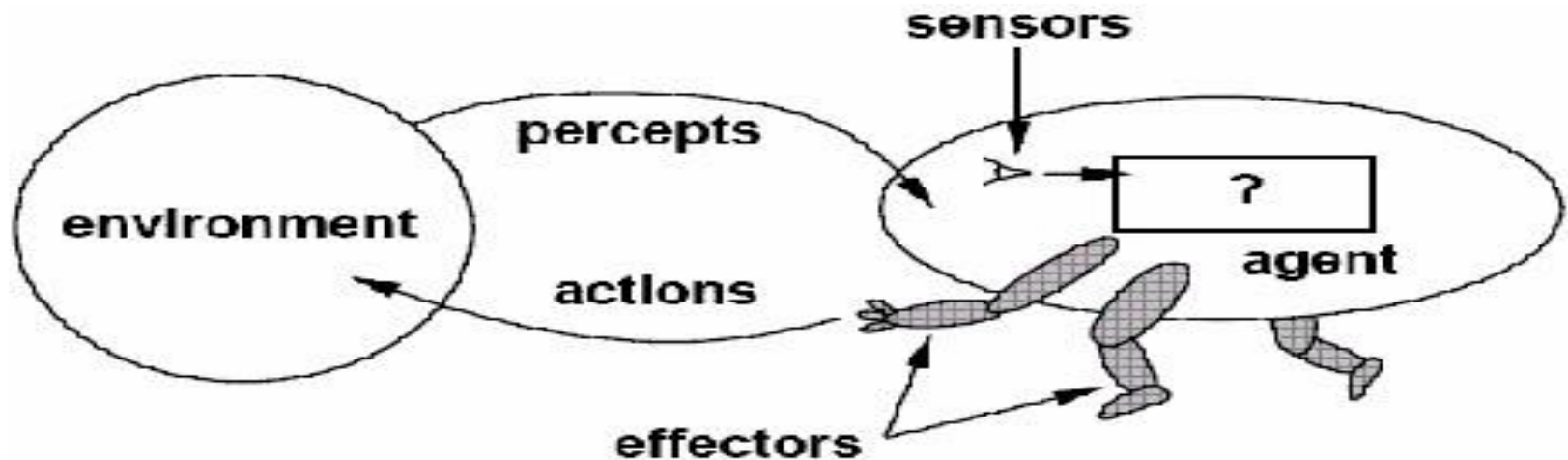
Chapter contents

- Agents and Environments.
- Rationality, omniscience, and autonomy in AI
- Task Environments in AI and properties of Task environments with PEAS examples.
- Structure of intelligent agents
- Intelligent Agent Types such as Simple reflex agent, Model-based reflex agent, Goal-based agent, Utility-based agent, Learning agent and their features.

Agent and Environment

- Artificial Intelligent system is composed of **Agent** and its **Environment**.
- An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators** to achieve its objectives.
- The **environment** is the external context in which an agent operates.

Agent and Environment cont....



Agent and Environment cont....

Main Components of an Intelligent Agent

- **Sensors:** Devices that detect changes in the environment and send information to other components.
- **Actuators:** Machine components that convert energy into motion, Actuators are responsible for moving and controlling a system. e,g motors and gears
- **Effectors:** The **devices that actually affect changes** in the environment through actions.
- Most AI agent has to follow the following four rules
 - ✓ **Rule1:**An AI agent must have the **ability to perceive** its environment.
 - ✓ **Rule2:** The **observations must be used to make decisions**.
 - ✓ **Rule3:** Each **decision should result in an action**.
 - ✓ **Rule4:**The **action taken must be rational**, leading toward the best outcome.

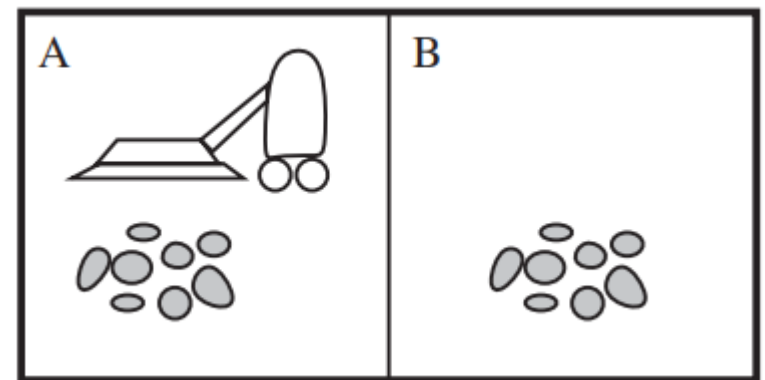
Agent and Environment cont....

Examples of Agents

Agent Type	Sensors	Actuators/Effectors	Example Actions
Human Agent	Eyes, ears, skin	Hands, legs, mouth	Speaking, walking, writing
Robotic Agent	Cameras, infrared sensors	Motors, gears	Moving, grasping objects
Software Agent	Keystrokes, file contents, network data	Display output, send packets	Managing emails, monitoring systems

Agent Function VS Agent Program

- The vacuum-cleaner world
 - has just two locations: squares A and B.
- The vacuum agent
 - perceives which square it is in and whether there is dirt in the square.
 - Can choose to move left, move right, suck up the dirt, or do nothing.
- Agent function
 - if the current square is dirty, then suck;
otherwise, move to the other square.



Agent Function VS Agent Program

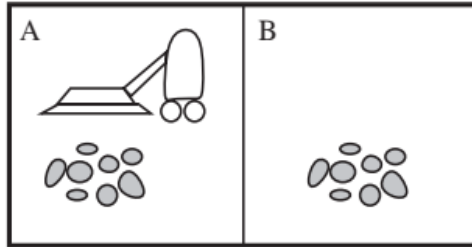


Figure 2.2 A vacuum-cleaner world with just two locations.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Agent function

```

function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
    
```

Figure 2.8 The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 2.3.

Agent program

Agent Function VS Agent Program

- **Agent function** –describes an agent's behavior by mapping any given **percept sequence** to an action.
- It is an **abstract mathematical description** that defines how an agent should act based on what it perceives.
 - Describes *what* an agent **should do** for every possible input
 - Can be represented (theoretically) as a **table** listing percept sequences and corresponding actions.
 - Represents an **external characterization** of an agent's behavior.
 - The table may be **very large** or even **infinite** in size.
- **Agent program**- The **internal implementation** of the agent function within an artificial agent. It is a **concrete algorithm or program** that runs on a physical machine.

The Concept of Rationality

- A **rational agent** is an agent that **acts to achieve the best possible outcome**, or when there is **uncertainty**, it acts to achieve the **best expected outcome**.
- Rationality depends on **four things**:
 - The **performance measure** that defines the criterion of success.
 - The agent's **prior knowledge** of the environment.
 - The **actions** that the agent can perform.
 - The agent's **percept sequence** to date.

Rationality

- This leads to a **definition of a rational agent**:
 - For each possible **percept sequence**, a rational agent should select **an action** that is expected to **maximize its performance measure**, given the evidence provided by the **percept sequence** and whatever **built-in knowledge** the agent has.

Design Performance Measure with Environment Focus

- It is better to design performance measures according to what one actually wants in the environment, rather than according to how one thinks the agent should behave.
- For example, the vacuum-cleaner agent, we propose to measure the performance by **the amount of dirt cleaned up in a single eight-hour shift.**

Omniscience

- **An omniscient agent** knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.
- Rational agents take action with expected success, where as omniscient agent take action with 100% sure of its success.

Rational vs Perfection

- Rationality maximizes **expected performance**, while perfection maximizes **actual performance**.
- Rationality does not require omniscience, then, because the rational choice depends only on the **percept sequence** *to date*.
- A rational agent requires **Information gathering** and **Exploration** to perform actions in order to modify future percepts so as to obtain useful information.

Learning and Autonomy

- An agent requires to **learn** as much as possible from what it perceives.
- The agent's initial configuration could reflect some prior knowledge of the environment, but as the agent gains experience this may be modified and augmented.
- There are extreme cases in which the environment is completely known **a priori**. In such cases, the agent need not perceive or learn; it simply acts correctly.
- To the extent that an agent relies on the prior knowledge of its designer rather than on its own percepts, we say that the agent lacks **autonomy**.
- A rational agent should be **autonomous**, it should learn what it can to compensate for **partial or incorrect prior knowledge** about the environment.

Specifying the Task Environment

- In designing an agent, the first step must always be to specify the **task environment** as fully as possible.
- **Task Environment** is the description Performance, Environment , Actuators, and Sensor(PEAS)
 - **Performance measure**
 - The performance measure that defines degrees of success of the agent
 - **Environment:** Knowledge of its surrounding
 - What an agent already knows about the environment it works in
 - **Actuators** – generating actions
 - The actions that the agent can perform back to the environment
 - **Sensors** – receiving percepts
 - Everything that the agent has perceived so far concerning the current scenario in the environment

Example Agent with their PEAS

Agent	Performance Measure	Environment	Actuators	Sensors
Vacuum Cleaner	Cleanness Efficiency Battery Life Security	Room, Table Wood floor Carpet, Various Obstacles	Wheels Brushes Vacuum Extractor	Camera Dirt detection sensor Cliff sensor Bump sensor Infrared wall sensor
Automated Car Drive	Comfortable trip Safety Maximum Distance	Roads Traffic Vehicles 	Steering wheel Accelerator Brake Mirror	Camera GPS Odometer
Hospital Management System	Patient's health Admission process Payment	Hospital Doctors Patients	Prescription Diagnosis Scan report	Symptoms Patient's response

Properties of Task Environments

- ❖ An **environment** is everything in the world which surrounds the agent.
- ❖ The environment is where agent lives, operate and provide the agent with something to sense and act upon it.
- ❖ Agent perceives from and acts upon an environment.
- ❖ Properties of Environments:
 - **Fully observable** (vs. **partially observable**):
 - **Deterministic** (vs. **stochastic**):
 - **Episodic** (vs. **sequential**):
 - **Static** (vs. **dynamic**):
 - **Discrete** (vs. **continuous**):
 - **Single agent** (vs. **multi-agent**):

Fully observable (vs. partially observable):

- Does the agent's sensory see the complete state of the environment?
 - If an agent has access to the complete state of the environment, then the **environment is accessible or fully observable.**
- An environment is effectively accessible if the sensors **detect all aspects** that are relevant to the choice of action.
- If the sensors provide only limited information about the environment, it is **partially observable.**
- **Example:-** Chess checker are fully observable
 - Automatic taxi driving is partially observable.

Deterministic (vs. stochastic):

- If an agent's current state and selected action can **completely determine the next state of the environment**, then such environment is called a ***deterministic*** environment.
- A ***stochastic*** environment is **random** in nature and **cannot be determined completely by an agent**.
- In a deterministic, fully observable environment, agent does not need to worry about uncertainty.
- Taxi driving is non-deterministic (i.e. stochastic) Traffic may delay arrival → the action move north may take longer than expected. Weather or accidents may change the outcome.

Episodic (vs. sequential):

- In an **episodic environment**, there is a series of **independent, one-shot actions**. The agent only needs the **current percept** to decide the next action — past actions have **no impact** on future ones.
- In a **sequential environment**, the agent's **current decision** can affect **future states and rewards**. Therefore, the agent requires **memory** of past percepts and actions to choose the next best move.
- In an **episodic task environment**, the agent's experience is divided into **atomic episodes**. In each episode:
 - The agent receives a percept.
 - The agent performs a single action.
 - The next episode does **not depend** on previous actions.
 - Many **classification** or **recognition tasks** are episodic.
- **Examples:**
- **Episodic:** A mushroom-picking robot or a cleaning robot that reacts only to current sensory input.
- **Sequential:** Taxi driving (since moving to a pickup point, dropping passengers, and planning the next trip depend on previous actions).
- **Chess: Sequential**, because each move affects future game states and possible strategies.

Static (vs. dynamic):

- If the environment can **change itself** while an agent is deliberating then such environment is called a **dynamic environment** else it is called a static environment.
- Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
- However for dynamic environment, agents need to keep looking at the world at each action.
- Taxi driving is an example of a dynamic environment whereas Crossword puzzles are an example of a static environment.

Discrete (vs. continuous):

- An environment is **discrete** if there are a **finite number of percepts and actions** available to the agent.
- If the percepts or actions can take on a **continuous range of values**, the environment is **continuous**.
- ❖ A chess game comes under discrete environment as there is a finite number of moves that can be performed.
- ❖ A self-driving car is an example of a continuous environment.
- ❖ Taxi driving is continuous - speed location are in a range of continuous values.
- ❖ Chess is discrete - there are a fixed number of possible moves on each item.

Single agent (vs. multi-agent):

- If an agent operates **by itself** in an environment, it is called a **single-agent environment**.
- When **two or more agents** interact in an environment, they form a **multi-agent environment**.
- For example, an agent solving a crossword puzzle by itself is clearly in a single-agent environment, whereas an agent playing chess is in a two-agent environment.

Structure of intelligent Agent

- We have talked about agents by describing *behavior*-- the action that is performed after any given sequence of percepts.
- The job of AI is to design an **agent program** that implements the **agent function**-- the mapping from percepts to actions.
- We assume this program will run on some sort of computing device with physical **sensors** and **actuators**—we call this the **architecture**.

$$agent = architecture + program$$

The Structure of Agents

- **Architecture:** The **physical machinery** or platform on which an agent executes.
- **Program:** The **implementation of the agent function** — the software that decides the agent's actions.

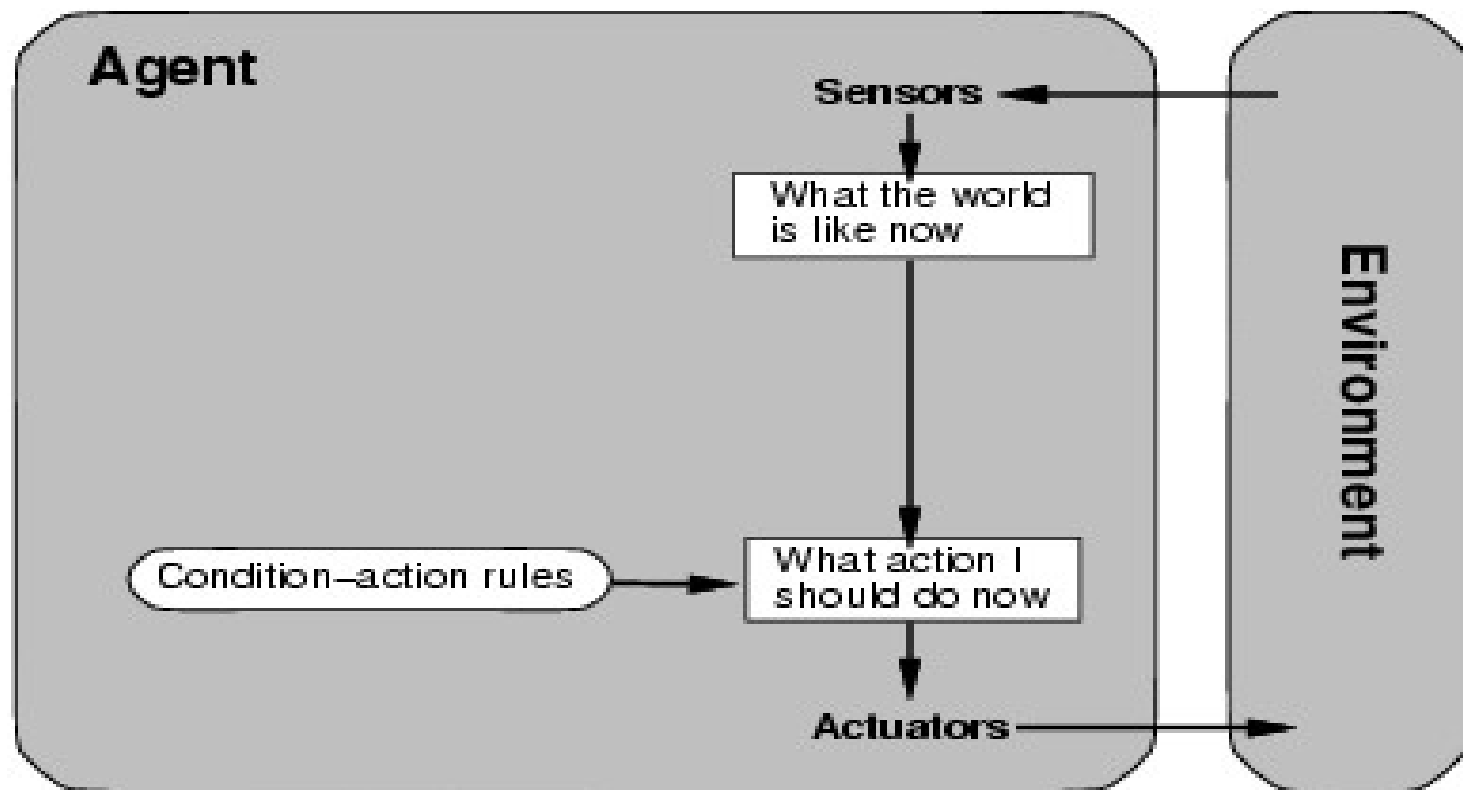
Types of Agents

- Agents can be grouped into five classes based on their **degree of perceived intelligence and capability**.
- All these agents can improve their performance and generate better action over the time.
 - **Simple reflex agents**
 - **Model-based reflex agents**
 - **Goal-based agents**
 - **Utility-based agents; and**
 - **Learning agents**
- Each kind of agent program combines particular components in particular ways to generate actions.

Simple reflex agents

- The simplest kind of agent is the **simple reflex agent**.
- These agents select actions on the basis of the **current percept**, ignoring the rest of the percept history.
- These agents only succeed in the fully observable environment.
- The Simple reflex agent **does not consider any part of percept history during their decision and action process**.
- Implemented through **condition-action rules (if- then rules)**
- For example, the vacuum agent is a simple reflex agent, because its decision is based only on the current location and on whether that contains dirt.

Simple reflex agents



Simple reflex agents

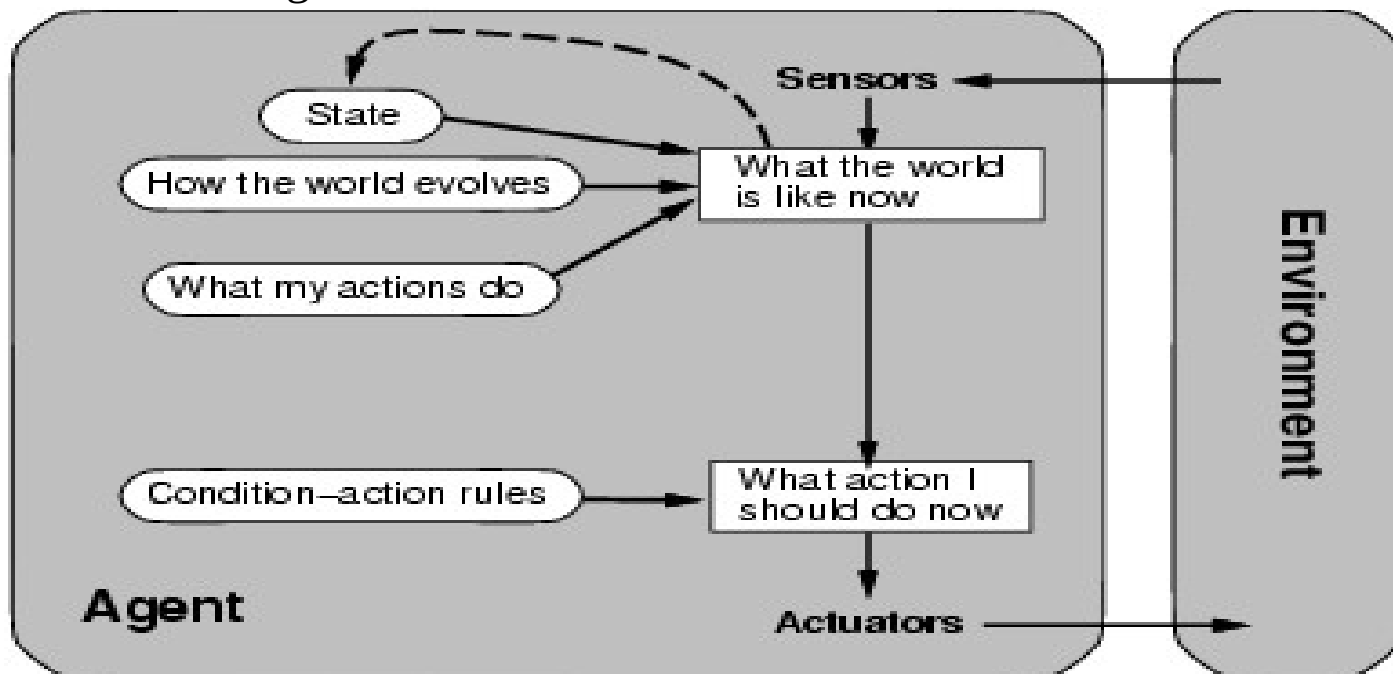
- Problems for the simple reflex agent design approach:
 - They have very limited intelligence
 - They do not have knowledge of non-perceptual parts of the current state
 - Operating in a partially observable environment, infinite loops are unavoidable

Model-Based Reflex Agent

- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
 - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
 - **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.

Model-Based Reflex Agent

- Updating the agent state requires information about:
 - How the world evolves
 - How the agent's action affects the world.

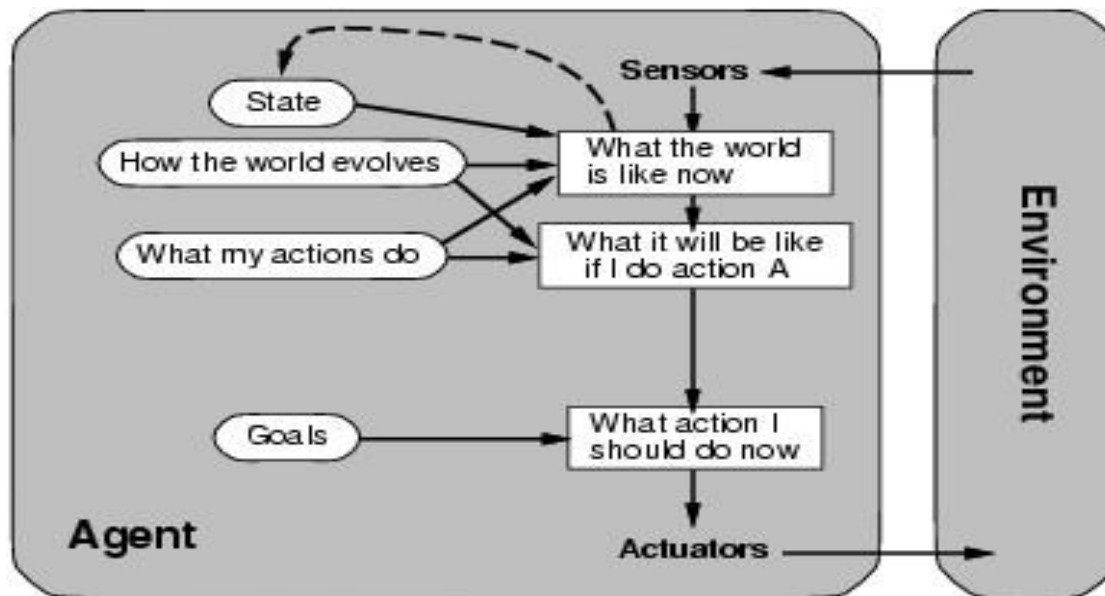


Goal-Based Agent

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its **goal** which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "**goal**" **information**.
- They choose an action, so that they can achieve the goal.

Goal-Based Agent

- These agents may have to consider a **long sequence of possible actions before deciding whether the goal is achieved or not**. Such considerations of different scenarios are called **searching** and **planning**, which makes an agent proactive.

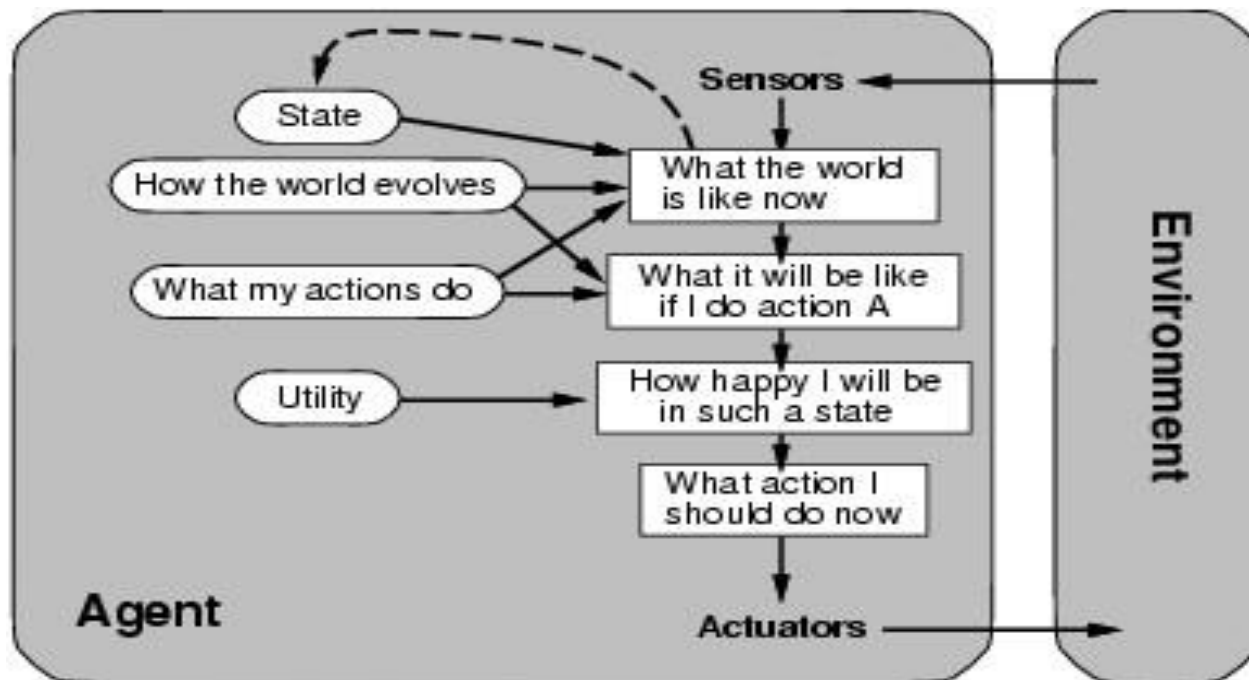


Utility-based Agent

- These agents are similar to the goal-based agent but provide **an extra component of utility measurement** which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.

Utility-based Agent

- The utility function maps each state to a real number to check how efficiently each action achieves the goals.



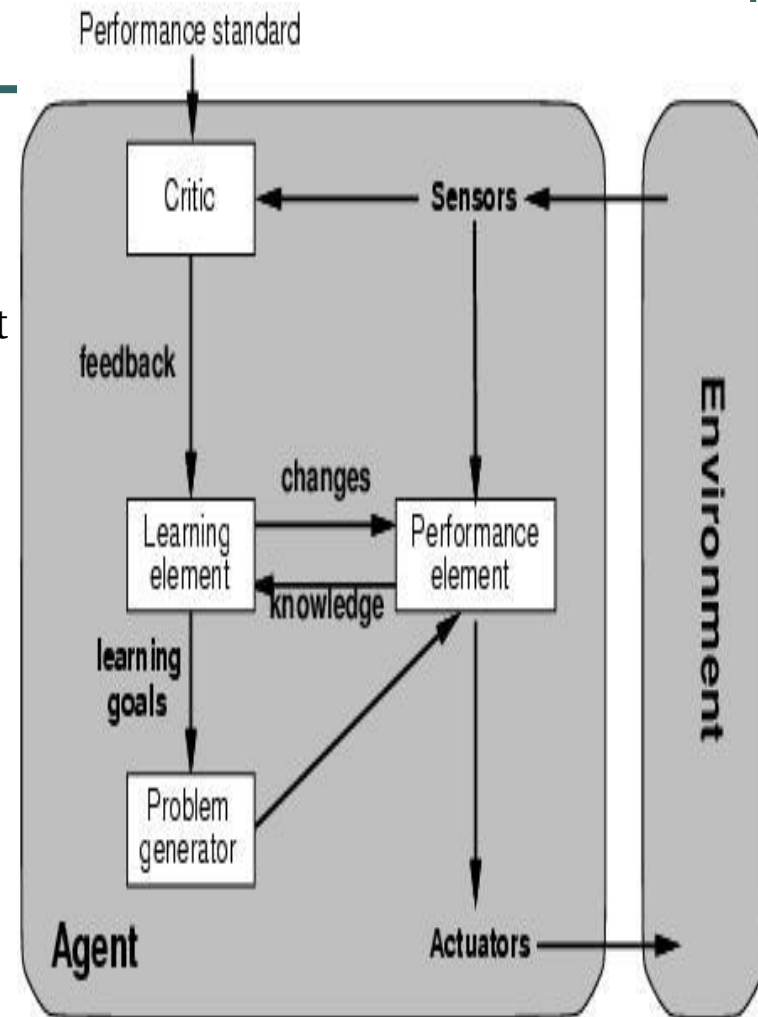
Learning Agent

- All agents can improve their performance through **learning**.
- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- A learning agent has mainly four conceptual components, which are:

Learning Agent

- **Learning element**:-Responsible for **making improvements** by learning from the environment.
- **Critic** -Provides **feedback** to the learning element about how well the agent is performing with respect to a **fixed performance standard**.
- **Performance element** -Responsible for **selecting and executing external actions** based on current knowledge.
- **Problem generator** -Suggests **new actions** that lead to **informative experiences**, enabling exploration and improvement.

Hence, learning agents are able to learn, analyse performance, and look for new ways to improve the performance.



Class Discussion

Task Environment	Observable	Deterministic	Episodic / Sequential	Static	Discrete	Agents
Crossword puzzle						
Taxi driver						
Tic-Tac-Toe						
Poker game						
Medical diagnosis system						
Online shopping recommender						
Traffic light controller						
Calculator program						

End of Chapter Two
THANKS!!!